

PAPER 1 / QUESTION 142

25 April 2024 / TZ2 (file label) / HL Paper 1 / original question 22 / 2224-6707

- 142.** A 1000 kg car accelerates from rest to a speed of 20 m s^{-1} . The car has an efficiency of $\frac{1}{3}$ and uses fuel of specific energy of 50 MJ kg^{-1} .

What mass of fuel is used to accelerate the car?

- A. 4 g
- B. 6 g
- C. 12 g
- D. 36 g